Notes: Augustin, Jiao, Sarkissian & Schill (RFS, 2020)
Cross-Listings and the Dynamics between Credit and Equity Returns

Contents

1 Big picture

- **Question.** Does listing a firm's equity in a foreign market change the relation between its CDS returns and stock returns?
- **Motivation.** Structural credit-risk models imply that credit and equity claims should be linked, but empirically the link is often weak.
- **Main result.** Cross-listing increases CDS-stock sensitivity, world market integration, and statistical synchronicity between CDS and stock prices.
- **Mechanism.** The evidence points mostly to reductions in information frictions: greater attention, visibility, analyst coverage, CDS coverage, and market familiarity.

2 Data and measurement

The sample covers 241 cross-listing events from 2001 to 2011, representing 215 firms with traded CDS contracts. The paper uses daily U.S. dollar-denominated five-year senior CDS contracts from IHS Markit, daily equity returns from Datastream and CRSP, analyst data from I/B/E/S, accounting data from Compustat Global, and measures of media attention, search intensity, liquidity, and governance.

Detailed interpretation: The data allow the authors to link an equity-market integration shock to credit-market outcomes.

3 Models and empirical specifications

3.1 Merton-style hedge-ratio intuition

$$h^E \approx \frac{\partial R^D}{\partial R^E}.$$

Debt and equity are contingent claims on the same firm value. If cross-listing reduces information frictions, the empirical hedge ratio should become larger in magnitude.

3.2 Main CDS-stock hedge-ratio regression

$$R_{i,t}^{CDS} = \alpha_i + \delta_t + \beta_1 R_{i,t}^E + \beta_2 CL_i R_{i,t}^E + \beta_3 R_{i,t-1}^E + \beta_4 CL_i R_{i,t-1}^E + \Gamma X_{i,t} + \varepsilon_{i,t}.$$

Detailed interpretation: The interaction terms capture whether CDS returns become more sensitive to equity returns after cross-listing.

3.3 Matched difference-in-differences

$$R_{i,t}^{CDS} = \cdots + \beta_2 D_i CL_i R_{i,t}^E + \beta_4 D_i CL_i R_{i,t-1}^E + \cdots.$$

Detailed interpretation: Matched non-cross-listed firms receive pseudo-event dates, helping distinguish treatment effects from market-wide trends.

4 Identification and empirical design

Cross-listing is not random, so the paper relies on event timing, firm and time fixed effects, controls, matched firms, and mechanism tests. The matched design is especially important because it asks whether similar non-cross-listed firms experience the same post-event increase in credit-equity integration. Mechanism tests then evaluate whether the effect is stronger where information frictions are most likely to matter.

5 Tables and figures

5.1 Figure 1: Correlation between CDS and stock returns around cross-listing

Read: The correlation is negative and becomes more negative after cross-listing, moving from about -0.12 to below -0.20.

Detailed interpretation: The figure motivates the paper by showing that CDS and stock prices become more synchronized after listing abroad.

Economic message: Cross-listing appears to make debt and equity prices move more consistently as claims on the same firm.

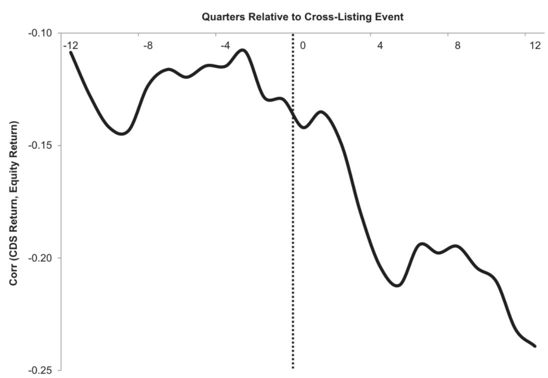


Figure 1
Correlation between CDS and stock returns around cross-listing
This figure shows the average quarterly Pearson correlation coefficient between daily CDS and stock returns around the cross-listing event. Each point on the plot represents the average correlation over three adjacent quarters ($t-1, t$, and $t+1$). The sample period is 2001–2013, and it includes 241 cross-listing events related to 278 CDS-stock pairs. CDS returns come from IHS Markit, and stock returns come from Datastream.

Frequency distribution of cross-listings by host and home markets

Home Host	ARE	AUT	AUS	BEL	BRA	CAN	CHL	COL	DNK	FIN	FRA	DEU	HKG	IND	ITA	JPN	LUX	MEX	NLD	NOR	PPL	PRT	SGP	ESP	SWE	SWZ	GBR	USA	Total
Arab Emirates																											1		1
Australia																							1				1	3	5
Belgium															1		2											2	5
Brazil													1															5	6
Canada					1		1	1											1									14	18
Chile																											1	1	2
China																											1	1	2
Colombia																												1	1
Czech Republic																					1								1
Finland															1				1										2
France				1											11	2	1		1							1	1	6	24
Germany		1													9				1							1		3	15
Greece			1																								1	1	3
Hong Kong																												1	1
Hungary																					1								1
Iceland																									1				1
India	1																14										3	2	20
Indonesia																	1												1
Ireland																											1	4	5
Italy											3	1																1	5
Japan											1		1														2	7	11
Kazakhstan																	2												2
Korea																1	5										1	3	10
Liechtenstein																										1			1
Luxembourg											2													1			2	1	6
Mexico																												2	2
Netherlands											2	1			4		1											5	13
New Zealand			2																										2
Norway																												1	1
Portugal																			1										1
Qatar																											1		1
Russia												1															2		3
Singapore													1				1											1	3
Spain															1				1			1							3
Sweden									1	1										1								1	4
Switzerland											1																	3	4
Taiwan																	10											1	11
Ukraine																												1	1
United Kingdom											2		1	1			1		1						1	1		15	23
United States						5					3					1		1	4							3	3		20
Total	1	1	3	1	1	5	1	1	1	1	14	3	4	1	27	4	38	1	11	1	2	1	1	1	2	7	21	86	241

This table provides the 2001–2011 distribution of cross-listed firms with traded CDS contracts for home and host markets shown with ISO-3 country codes. Firms with cross-listings from the Sarkissian and Schill public database are matched with the IHS Markit CDS data to ensure the availability of CDS contracts.

5.2 Table 1: Frequency distribution of cross-listings by host and home markets

Read: The sample has 241 events across 40 home and 28 host markets, with the United States the largest host market.

Detailed interpretation: The sample is broad and global, but selected toward firms with traded CDS contracts.

Economic message: The paper studies global firms whose equity and CDS are both traded.

5.3 Table 2: Chronology of CDS initiation relative to cross-listing events

Read: CDS trading starts before listing for some firms and after listing for others.

Detailed interpretation: Timing determines which firms can be used in symmetric pre/post tests.

Economic message: The paper balances sample size against clean event-window identification.

Table 2
Chronology of CDS initiation relative to cross-listing events

	CDS initiation month relative to the cross-listing month			
Home market	Before 3 months	Within ± 3 months	After 3 months	Total
Arab Emirates			1	1
Australia	3	1	1	5
Belgium	3		5	8
Brazil	2		4	6
Canada	8	1	11	20
Chile			2	2
China			2	2
Colombia			1	1
Czech Republic	1			1
Finland	1		1	2
France	15	5	9	29
Germany	12	2	2	16
Greece	1		2	3
Hong Kong			1	1
Hungary			1	1
Iceland			2	2
India	5	1	14	20
Indonesia			1	1
Ireland	1	1	5	7
Italy	3	2	4	9
Japan	6	1	4	11
Kazakhstan	2			2
Korea	4		6	10
Liechtenstein			1	1
Luxembourg	2	2	3	7
Mexico	1		1	2
Netherlands	13		3	16
New Zealand		1	1	2
Norway		1	1	2
Portugal	1			1
Qatar	1			1
Russia	2		1	3
Singapore	2		2	4
Spain	2		1	3
Sweden	2		2	4
Switzerland	1	1	4	6
Taiwan	6	2	3	11

atistics of firm characteristics by home market

e statistics of sample of daily CDS and equity returns

		CDS characteristic									
		Spread (%)		Coverage (#)		Return (%)		Equity return (%)			
t	Obs.	Mean	SD	Mean	SD	Mean	SD	Mean	SD	ρ	Stale
le (241 firms)	369,571	1.639	4.143	5.801	3.224	0.010	5.391	0.005	2.750	−0.14	0.17
ple (79 firms)	169,788	1.308	2.562	6.179	3.188	0.012	4.994	0.006	2.928	−0.15	0.17

e statistics of annual firm characteristics

		MkCap ($bln)		ROA		Leverage		P/B		Analysts (#)	
t	Obs.	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
le (241 firms)		28.73	19.14	0.039	0.070	0.243	0.151	2.620	6.451	20.8	11.3
ple (79 firms)		41.88	41.48	0.036	0.067	0.236	0.150	1.516	5.620	21.9	11.5

ows the mean and standard deviation of firm CDS spread levels, coverage, returns, and equity returns (panel A) and of five other characteristics (panel B) of firms with and traded CDSs. The sample period is 2001–2013. The overall sample contains 241 firms with available stock and CDS returns. The reduced sample contains only firms 1 year of stock and CDS returns before and after the cross-listing event, and there are 79 such firms. *CDS spread* is the firm's 5-year CDS spread. *CDS coverage* denotes the ealer quotes used in the computation of the 5-year mid-market spread. Both these variables come from IHS Markit. *Equity returns* (as a percentage) come from Datastream. son correlation coefficient between equity and *CDS returns* (as a percentage). Stale is the proportion of stale quotes (i.e., no change in spreads) among 5-year CDS contracts. ial accounting information comes from Compustat Global. *MkCap* is the market capitalization in billion U.S. dollars; *ROA* is the return on assets; *Leverage* is the long-term by the sum of long-term debt and market value of equity; and P/B is the price-to-book ratio. *Analysts* is the number of analysts covering a firm from I/B/E/S database.

5.4 Table 3: Summary statistics of firm characteristics by home market

Read: The sample has negative average CDS-stock correlation and substantial cross-country heterogeneity.

Detailed interpretation: The weak baseline correlation motivates the paper's question.

Economic message: Cross-listing studies firms whose credit and equity claims are not initially tightly synchronized.

5.5 Tables 4–5 and Figure 2: Hedge-ratio evidence

Read: CDS returns become more sensitive to equity returns after cross-listing, and this result is unique to treated firms relative to matched controls.

Detailed interpretation: These are the main treatment-effect tables. They show that cross-listing changes credit-equity pricing dynamics.

Economic message: Cross-listing strengthens the capital-structure linkage between equity and credit markets.

Table 4
Impact of cross-listing on CDS and stock return hedge ratios

Variable	(1)	(2)	(3)	(4)	(5)
$R^E_{i,t}$	−0.127***	−0.055***	−0.121***	−0.093***	−0.103***
	(4.13)	(2.87)	(4.12)	(3.33)	(3.26)
$CL_i \times R^E_{i,t}$	−0.205***	−0.155***	−0.209***	−0.224***	−0.210***
	(5.61)	(5.57)	(5.92)	(6.58)	(5.46)
$R^E_{i,t-1}$	−0.090***	−0.084***	−0.093***	−0.078***	−0.085***
	(4.61)	(4.39)	(4.71)	(3.69)	(3.54)
$CL_i \times R^E_{i,t-1}$	−0.039**	−0.040**	−0.042**	−0.067***	−0.063***
	(2.00)	(2.06)	(2.12)	(3.27)	(2.64)
CL_i	0.075***	0.068***	0.066**	0.098*	0.072
	(4.09)	(3.74)	(2.15)	(1.72)	(1.39)
$R^{CDS}_{i,t-1}$	−0.045***	−0.045***	−0.058***	−0.121***	−0.133***
	(4.41)	(4.38)	(5.99)	(10.01)	(10.11)
$Leverage_{i,t}$		−0.104	−0.192	−0.194	−0.426
		(0.88)	(1.31)	(0.78)	(1.53)
$EqVol_{i,t}$		0.049***	0.078***	0.096***	0.088***
		(5.67)	(8.10)	(6.56)	(5.65)
$ROA_{i,t}$		0.271*	0.256	0.453*	−0.194
		(1.71)	(1.26)	(1.95)	(0.51)
$P/B_{i,t}$		0.115	0.766	0.113	−0.913
		(0.74)	(0.90)	(1.00)	(1.55)
$R_{c,t}$		−0.261***	−0.260***	−0.244***	
		(10.09)	(10.24)	(10.35)	
$R_{w,t}$		−0.629***			
		(19.59)			
ΔVIX_t		−0.012***			
		(4.68)			
Time FEs			Yes	Yes	Yes
Host × Time FEs				Yes	
Home × Time FEs					Yes
Obs.	367,085	367,023	366,887	340,547	323,188
R^2	.036	.055	.087	.267	.279

This table shows the impact of cross-listings on the hedge ratio of a firm's CDS and stock returns. The sample period is 2001–2013, and the data are daily. The dependent variable is the CDS return, $R^{CDS}_{i,t} \cdot R^E_{i,t}$ is the gross equity return of firm i at date t. CL_i is a dummy variable, which equals 1 after firm i cross-lists and 0 otherwise. $R_{w,t}$ is the MSCI world index return, $R_{c,t}$ contains the residuals from a regression of the home market MSCI country index returns on the world index returns. *Leverage* is the long-term debt, divided by the sum of long-term debt and market value of equity; *EqVol* is the equity volatility; *ROA* is the return on assets; P/B is the price-to-book ratio; and *VIX* is the CBOE volatility index. CDS returns come from IHS Markit. Firm and equity market returns come from Datastream. Firms' accounting information comes from Compustat Global. Each regression includes firm fixed effects and a constant, for which the estimated coefficient are not shown. Most regressions also contain daily time fixed effects. Home (Host) × Time FEs denotes the interaction of home (host) country and time fixed effects. The standard errors are clustered by firm and by date. The absolute t-statistics are in parentheses. The number of observations and the adjusted R^2 are also reported. $^*p<.1$; $^{**}p<.05$; $^{***}p<.01$.

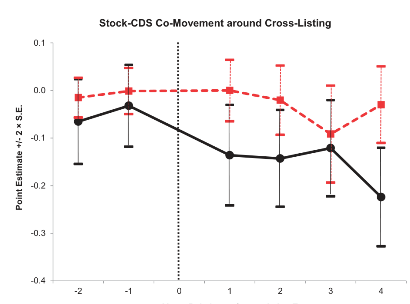
Table 4: Impact of cross-listing on CDS and stock return hedge ratios

Table 5
Impact of cross-listing on CDS and stock return hedge ratios for cross-listed and matched firms

	Matched	Cross-listed	DiD				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$R^E_{i,t}$	−0.124***	−0.127***	−0.123***	−0.066***	−0.111***	−0.090***	−0.069***
	(6.96)	(4.13)	(6.97)	(3.63)	(6.53)	(6.58)	(4.50)
$CL_i \times R^E_{i,t}$	−0.029	−0.205***	−0.029	−0.025	−0.029	−0.055	−0.087**
	(0.66)	(5.61)	(0.66)	(0.80)	(0.72)	(1.26)	(2.36)
$R^E_{i,t-1}$	−0.101***	−0.090***	−0.099***	−0.062***	−0.094***	−0.0795***	−0.063***
	(7.70)	(4.61)	(7.71)	(4.27)	(7.13)	(6.06)	(3.83)
$CL_i \times R^E_{i,t-1}$	−0.009	−0.039**	−0.079	−0.023	−0.012	−0.029	−0.059**
	(0.32)	(2.00)	(0.30)	(0.92)	(0.45)	(0.96)	(2.06)
CL_i	0.058***	0.075***	0.067***	0.069***	0.075***	0.069**	0.128***
	(3.33)	(4.09)	(5.68)	(5.90)	(3.84)	(1.99)	(2.73)
$R^{CDS}_{i,t-1}$	−0.078***	−0.045***	−0.058***	−0.058***	−0.067***	−0.146***	−0.145***
	(5.62)	(4.41)	(6.64)	(6.57)	(7.68)	(12.35)	(12.67)
$D_i \times R^E_{i,t}$			−0.040	0.006	−0.009	−0.011	−0.033
			(0.09)	(0.20)	(0.21)	(0.30)	(1.00)
$D_i \times R^E_{i,t-1}$			0.794	−0.025	0.021	−0.099	−0.021
			(0.28)	(0.88)	(0.01)	(0.36)	(0.73)
$D_i \times CL_i \times R^E_{i,t}$			−0.176***	−0.134***	−0.186***	−0.187***	−0.146***
			(2.77)	(2.94)	(3.10)	(3.04)	(2.74)
$D_i \times CL_i \times R^E_{i,t-1}$			−0.034	−0.020	−0.035	−0.042	−0.016
			(0.94)	(0.57)	(0.98)	(1.07)	(0.42)
Controls				Yes	Yes	Yes	Yes
Time FEs					Yes	Yes	Yes
Home × Time FEs						Yes	
Host × Time FEs							Yes
Obs.	286,986	367,085	654,070	654,070	653,983	551,209	566,015
R^2	.017	.036	.027	.045	.060	.265	.258

This table shows the impact of cross-listing on the sensitivity of a firm's CDS and stock returns for cross-listed firms and matched non-cross-listed firms. The sample period is 2001–2013. The matched sample is constructed using a Euclidean distance-based matching algorithm, as described in Table A.3 of the Internet Appendix. Matched firms use a pseudo-cross-listing date identical to the corresponding cross-listed firm. The dependent variable is the daily CDS return, $R^E_{i,t-1} \cdot R^E_{i,t}$ is the gross equity return for firm i at date t. CL_i is a dummy variable, which equals 1 after firm i cross-lists and 0 otherwise. D_i is a dummy variable that equals 1 for a cross-listed firm and 0 for a matched firm. All control variables are the same as those used in Table 4, but their estimates are not reported. CDS returns come from IHS Markit. Firm and market equity returns come from Datastream and CRSP, respectively. Firms' financial accounting information comes from Compustat Global. DiD are the estimates of the differences-in-differences tests. Each regression includes firm fixed effects and a constant, for which the estimated coefficients are not shown. Some regressions also contain daily time fixed effects. Home (Host) × Time FEs is the interaction of home (host) country and daily time fixed effects. The standard errors are clustered by firm and time. The absolute t-statistics are in parentheses. The number of observations and the adjusted R^2 are also reported. $^*p<.1$; $^{**}p<.05$; $^{***}p<.01$.

(a) Matched-firm hedge-ratio results



(b) Hedge-ratio dynamics

5.6 Table 6: World market integration test results before and after cross-listing events

Read: CDS and equity returns become more exposed to global equity and bond market factors.

Detailed interpretation: The result extends the evidence from firm-level CDS-stock sensitivity to global factor integration.

Economic message: Cross-listing integrates multiple claims on the same firm into world capital markets.

Table 6
World market integration test results before and after cross-listing events

	IM-1		IM-2		IM-3		IM-4	
	Equity	Credit	Equity	Credit	Equity	Credit	Equity	Credit
β_w^{stock}	1.075	−0.579	1.078	−0.577	1.072	−0.558	0.965	−0.837
$\Delta\beta_w^{stock}$	0.007	−0.247**	−0.002	−0.262**	0.003	−0.262**	0.076	−0.331**
β_c^{stock}	0.835	−0.120	0.833	−0.141	0.836	−0.126	0.859	−0.202
$\Delta\beta_c^{stock}$	−0.037	−0.144	−0.034	−0.144*	−0.039	−0.142	−0.081	−0.240*
β_w^{bond}	0.489	0.129	0.480	0.189	0.489	0.236	0.519	0.388
$\Delta\beta_w^{bond}$	0.178**	0.464**	0.182**	0.333*	0.177**	0.353*	0.284**	0.391**
β_c^{bond}	−0.575	0.285	−0.590	0.397	−0.562	0.398	−0.736	0.394
$\Delta\beta_c^{bond}$	−0.249*	0.834**	−0.221	1.010***	−0.259*	1.059***	−0.125	1.013**
β_{RX}			−0.001	−0.022				
$\Delta\beta_{RX}$			0.029	0.063				
β_{IM}			−0.004	−0.079				
$\Delta\beta_{IM}$			0.007	−0.054				
β_{IM}					0.003	−0.056		
$\Delta\beta_{IM}$					0.008*	0.003		
β_{smb_g}							−0.039	−0.969
$\Delta\beta_{smb\_g}$							0.105	−0.406
β_{smb_l}							0.113	−0.148
$\Delta\beta_{smb\_l}$							−0.084	−0.092
β_{hml_g}							0.195	−0.585
$\Delta\beta_{hml\_g}$							0.277	0.109
β_{hml_l}							0.312	0.233
$\Delta\beta_{hml\_l}$							−0.016	−0.395

This table shows world market integration tests for equity and CDS returns around cross-listing using integration model IM-1 and its extensions. The sample period is 2001–2013. The dependent variable is the firm's stock or CDS return in excess of the 1-month Treasury bill. The CDS return data come from Markit. The stock and bond market data come from Datastream. All returns are denominated in U.S. dollars. $\beta_w^{stock} \cdot (\beta_w^{bond})$ is the world stock (bond) market beta, $\beta_c^{stock} \cdot (\beta_c^{bond})$ is country stock (bond) beta before cross-listing. $\Delta\beta$ shows the average difference between the betas before and after cross-listings and its level of significance. Integration models IM-2, IM-3, and IM-4 also report the betas of CDS and stock returns to currency factors (dollar risk, β_{RX}, and carry trade, β_{FX}), the intermediary capital factor, β_{IM}, and the Fama-French global and domestic HML and SMB factors, receptively. The estimation is for each firm 1 year before and 1 year after cross-listing, excluding the event day. Only firms with at least 1 year of stock and CDS returns before and after cross-listing are included in estimations, and there are 79 such firms. The two stock market factors are the excess returns on the world market equity index (*MSCI World*) and a firm's domestic equity index, orthogonal to the world market. The two bond factors are the Citigroup World Government Bond Index return in U.S. dollars and the Citigroup Government Bond Index return in each home market orthogonal to the first index. The currency factors come from Lustig, Roussanov, and Verdelhan (2011). The intermediary capital factor comes from He, Kelly, and Manela (2017). The Fama-French domestic *HML* (*SMB*) factor is the regional *HML* (*SMB*) factor orthogonal to the global *HML* (*SMB*) factor. $^*p<.1$; $^{**}p<.05$; $^{***}p<.01$.

Table 6: World market integration test results

5.7 Table 7, Figure 3, and Table 8: Price synchronicity

Read: Pricing discrepancies fall and statistical integration rises after cross-listing.

Detailed interpretation: These tests show the result is not only a linear hedge-ratio result; CDS and stock prices become more synchronized nonparametrically.

Economic message: Credit-equity integration improves in several distinct empirical senses.

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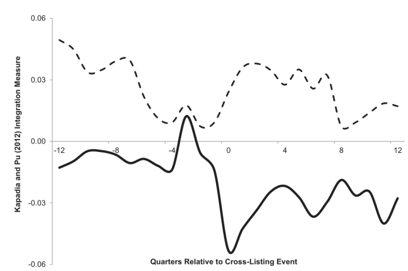
	Interval (τ)	Before cross-listing Mean	Before cross-listing SD	After cross-listing Mean	After cross-listing SD	Diff (after – before)	t-stat
A. Pricing discrepancy measure for the overall sample of firms							
$I(R^{CDS} \times R^E < 0)$	1	0.337	0.148	0.509	0.122	0.171***	7.88
	5	0.487	0.131	0.600	0.104	0.119***	5.97
	10	0.553	0.136	0.610	0.102	0.057***	2.98
	20	0.558	0.186	0.616	0.171	0.058**	2.03
$I(R^{CDS} \times R^E = 0)$	1	0.385	0.251	0.136	0.166	−0.249***	7.37
	5	0.132	0.170	0.045	0.105	−0.087***	2.62
	10	0.070	0.131	0.028	0.091	−0.041**	2.33
	20	0.075	0.114	0.048	0.175	−0.027	1.14
$I(R^{CDS} \times R^E > 0)$	1	0.276	0.111	0.355	0.067	0.079***	5.38
	5	0.378	0.084	0.354	0.068	−0.023*	1.91
	10	0.373	0.100	0.361	0.081	−0.012	0.85
	20	0.367	0.128	0.336	0.101	−0.031*	1.70
B. Kendall's tau measure for a reduced sample of cross-listed and matched firms							
Cross-listing sample	1	−0.055	0.131	−0.161	0.112	−0.106***	6.29
	5	−0.096	0.150	−0.225	0.166	−0.131***	6.78
	10	−0.152	0.215	−0.260	0.163	−0.108***	4.14
	20	−0.183	0.212	−0.297	0.171	−0.114***	4.21
Matched sample	1	−0.051	0.081	−0.100	0.082	−0.049***	3.76
	5	−0.073	0.123	−0.143	0.126	−0.070***	4.41
	10	−0.074	0.224	−0.159	0.149	−0.085***	3.19
	20	−0.112	0.238	−0.194	0.213	−0.081***	2.68
DiD test	1					−0.057***	3.06
	5					−0.061**	2.34
	10					−0.023	0.63
	20					−0.033	0.53

This table reports the price synchronicity results of Kapadia and Pu (2012) before and after cross-listing. The sample period is 2001–2013. Panel A shows the means and standard deviations of pricing discrepancies between stock prices and CDS spreads; panel B provides the same statistics for Kendall's tau for the cross-listed and matched firms, using only firms with at least 1 year of stock and CDS returns before and after cross-listing. There are 79 such firms. The pricing discrepancy measure depends on the concordance of CDS spread and stock returns: $I(R^{CDS} \times R^E > 0), I(R^{CDS} \times R^E = 0)$, or $I(R^{CDS} \times R^E < 0)$. Here, I is an indicator function, R_i^{CDS} and R_i^E are the CDS spread and stock returns, computed using the logarithmic changes in stock prices, P_i^E, and CDS spreads, P_i^{CDS}, of firm i: $R_{i,k}^{E,\tau} = \ln(P_i^E(k+\tau)/P_i^E(k)), R_{i,k}^{CDS,\tau} = \ln(P_i^{CDS}(k+\tau)/P_i^{CDS}(k))$; $\tau = [1, 5, 10, 20]$ corresponds to the return horizon in days. The equity and CDS markets are aligned if $I(R^{CDS} \times R^E < 0)$; they are neither aligned nor misaligned if $I(R^{CDS} \times R^E = 0)$; and they are misaligned if $I(R^{CDS} \times R^E > 0)$. All pricing discrepancy measures are computed over nonoverlapping time intervals. Kendall's tau correlation is computed as $K_i = 4\kappa_i \,/\, (\mathrm{M}(\mathrm{M}-1)) - 1$, where κ_i is the integration measure:

$$\kappa_i = \sum_{k=1}^{M-\tau} I\left[R_{i,k}^{CDS,\tau} \times R_{i,k}^{E,\tau} > 0\right].$$

(a) Kapadia-Pu price discrepancies



(b) Integration dynamics

Table 8
Credit-equity integration before and after cross-listing for cross-listed and matched firms

	(1)	(2)	(3)	(4)	(5)	(6)
CL_i	0.026**	0.027**	0.026**	0.024*	0.024*	0.026**
	(2.04)	(2.15)	(2.05)	(1.91)	(1.91)	(2.05)
$D_i \times CL_i$	−0.049***	−0.051***	−0.049***	−0.043**	−0.047***	−0.048***
	(2.84)	(3.00)	(2.87)	(2.53)	(2.76)	(2.85)
Leverage	−0.023	−0.018	−0.024	−0.018	−0.019	−0.021
	(0.63)	(0.50)	(0.66)	(0.50)	(0.52)	(0.57)
EqVol	0.011	0.015	0.011	0.067	0.053	−1.280
	(0.15)	(0.20)	(0.15)	(0.92)	(0.07)	(1.48)
MkCap	0.060	0.069	0.059	0.010	0.065	0.069
	(0.91)	(1.09)	(0.90)	(0.14)	(0.93)	(1.06)
ZeroSpread	−0.013					
	(0.09)					
Depth		−0.070***				
		(5.41)				
ZeroRet			−0.028			
			(0.66)			
Illiquidity				0.011***		
				(3.67)		
TrVolume					0.034	
					(1.27)	
IVol						1.385
						(1.60)
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes
Time FEs	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	8,850	8,850	8,850	8,712	8,712	8,850
R^2	.431	.435	.431	.440	.437	.434

This table reports the results of the panel regression:

$$\bar{K}_{i,t} = a_0 + b_1 CL_i + b_2 D_i \times CL_i + c_1' X_{i,t} + \alpha_i + \delta_t + \varepsilon_{i,t},$$

Table 8: Credit-equity integration before and after cross-listing

5.8 Tables 9–10: Mechanism tests

Read: Information-related variables such as analyst coverage, CDS coverage, media attention, search intensity, and familiarity explain where the treatment effect is strongest.

Detailed interpretation: The horse-race evidence favors information frictions over governance as the main mechanism.

Economic message: Cross-listing improves price alignment mostly by increasing attention and information production.

Table 9
Impact of cross-listing on CDS and stock returns hedge ratios by firm and country characteristics

		High-low (level)		High-low (change)	
		Coefficient	t-stat	Coefficient	t-stat
A. Firm characteristics					
A.1 Trading frictions	*EqVol*	−0.248***	(4.65)	−0.022	(0.34)
	Leverage	−0.070	(1.17)	−0.076	(1.09)
	IVol	−0.180***	(2.75)	−0.010	(0.16)
	Illiquidity	−0.119*	(1.94)	−0.104*	(1.70)
A.2 Information frictions	*TrVolume*	−0.093	(1.46)	−0.159**	(2.53)
	Analyst coverage	−0.242***	(3.85)	−0.189***	(3.19)
	CDS coverage	−0.294***	(4.88)	−0.197***	(3.24)
(attention/visibility)	*Advertising*	−0.115	(1.62)	−0.029	(0.31)
	Media	−0.354***	(5.52)	−0.042	(0.65)
	Search	−0.190**	(2.25)	−0.025	(0.33)
A.3 Governance	*Tangibility*	−0.022	(0.37)	−0.039	(0.61)
	Collateral	−0.077	(0.01)	−0.033	(0.49)
B. Country characteristics					
B.1 Familiarity	*Geographic proximity*	−0.243***	(4.13)		
	Cross-country correlation	−0.242***	(3.08)		
B.2 Governance	*Rule of law*	−0.098	(1.39)		
	Disclosure requirement	−0.010	(0.31)		

Table 10
Comparison between trading cost and information cost characteristics

A. High-low (level)

		EqVol	*IVol*	*Amihud*
Analyst coverage	$D_T \times CL_i \times R^E_{i,t}$	0.188**	−0.035	0.014
	$D_I \times CL_i \times R^E_{i,t}$	−0.169**	−0.352***	−0.270***
	$D_T \times D_I \times CL_i \times R^E_{i,t}$	0.164*	0.332**	0.030
CDS coverage	$D_T \times CL_i \times R^E_{i,t}$	0.050	0.024	−0.076
	$D_I \times CL_i \times R^E_{i,t}$	−0.226***	−0.163*	−0.269***
	$D_T \times D_I \times CL_i \times R^E_{i,t}$	0.380***	0.093	0.211*
Media	$D_T \times CL_i \times R^E_{i,t}$	0.171**	−0.156	0.247***
	$D_I \times CL_i \times R^E_{i,t}$	−0.199***	−0.456***	−0.194**
	$D_T \times D_I \times CL_i \times R^E_{i,t}$	0.111	0.394***	0.257**
Search	$D_T \times CL_i \times R^E_{i,t}$	0.142	0.047	0.148
	$D_I \times CL_i \times R^E_{i,t}$	0.275***	0.128	0.135
	$D_T \times D_I \times CL_i \times R^E_{i,t}$	−0.118	0.107	−0.187

B. High-low (change)

		Amihud	*Trading volume*
Analyst coverage	$D_T \times CL_i \times R^E_{i,t}$	0.035	−0.134
	$D_I \times CL_i \times R^E_{i,t}$	−0.106	−0.140
	$D_T \times D_I \times CL_i \times R^E_{i,t}$	−0.223*	−0.048
CDS coverage	$D_T \times CL_i \times R^E_{i,t}$	−0.063	0.058
	$D_I \times CL_i \times R^E_{i,t}$	−0.251***	−0.188**
	$D_T \times D_I \times CL_i \times R^E_{i,t}$	0.259**	0.035

This table shows the impact of cross-listing on the sensitivity of a firm's CDS and stock returns. We split all firms into "High" and "Low" subsamples, based on the median value of the characteristic 1 year before the cross-listing event (panel A). Alternatively, we split firms into "High" and "Low" subsamples, based on the median change of the characteristic from 1 year before to 1 year after the cross-listing event (panel B). The daily CDS return, $R^E_{i,t}, R^E_{i,t-1}$, is regressed on daily stock returns: $R^E_{i,t}, R^E_{i,t-1}$, and their interactions with the cross-listing dummy,

5.9 Table 11 and Figure 4: Friction and governance indices

Read: Trading and information friction indices fall after cross-listing, while governance does not show a comparable improvement.

Detailed interpretation: These results validate the mechanism variables directly.

Economic message: The strongest mechanism is reduced information frictions, not governance bonding.

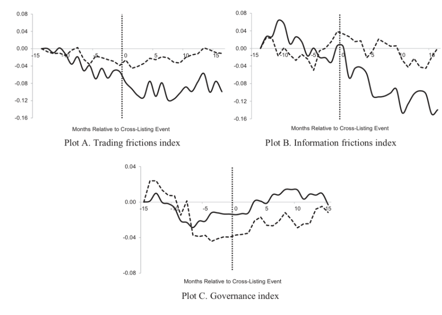
d governance indices before and after cross-listing for cross-listed and matched firms

Trading frictions index			Information frictions index			Governance index		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
−0.004	−0.004	0.224	−0.085	−0.097*	0.136**	−0.014	−0.017	−0.060**
(0.29)	(0.32)	(1.67)	(1.30)	(1.88)	(1.99)	(0.32)	(0.54)	(2.15)
−0.202***	−0.198***	−0.215***	−0.189***	−0.178***	−0.211***	0.005	0.003	−0.011
(5.39)	(5.08)	(6.05)	(2.82)	(2.95)	(3.96)	(0.19)	(0.09)	(0.29)
No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
No	No	Yes	No	No	Yes	No	No	Yes
3,636	3,636	3,634	3,636	3,636	3,634	3,595	3,595	3,590
.057	.645	.696	.045	.804	.867	.001	.977	.979

ɔorts the results of the panel regression:

$$X_{i,t} = a_0 + b_1 CL_i + b_2 D_i \times CL_i + \alpha_i + \delta_t + \varepsilon_{i,t},$$

s the trading frictions index, the information frictions index, or the governance index. CL_i is a dummy variable, which equals 1 after firm i cross-lists and 0 otherwise. my variable that equals 1 for a cross-listed firm and 0 for a matched firm. The sample period is 2001–2013. The estimation window covers [−15, −3] and [+3, +15] nd (pseudo-)cross-listing events. All variables are measured at the monthly frequency. Firm α_i and (monthly) time fixed effects δ_t are included in some specifications. s (1), (4), and (7) also include D_i. Each regression specification includes a constant, for which the estimate is not reported in the results. The standard errors are firm and by time. The absolute t-statistics are reported in parentheses. The number of observations and the adjusted R^2 are also reported. *$p < .1$; **$p < .05$; ***$p < .01$.



(a) Friction and governance index regressions

(b) Friction dynamics

6 Short summary

- Cross-listing increases CDS-stock sensitivity, world market integration, and CDS-stock synchronicity.
- Matched-firm tests show the effect is unique to cross-listed firms.
- Mechanism tests point mainly to information frictions and investor attention.
- Cross-listing has spillovers beyond equity markets and affects credit pricing.

7 Conclusion

7.1 Core conclusion

Foreign equity cross-listings change the pricing dynamics of firms' credit claims.

7.2 Economic intuition

Cross-listing raises firm visibility, improves information production, and attracts arbitrage activity across claims.

7.3 Mechanism interpretation

The evidence favors reduced information frictions over governance improvements.

7.4 Limitations

Cross-listing is not randomly assigned, and the sample is limited to firms with traded CDS contracts.

7.5 Takeaway

Cross-listing helps integrate a firm's capital structure across equity and credit markets.